Bayesian Data Analysis of the German Tank Problem

Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Inferring Discrete Ceilings, Decision Theory, and Operational Assumption Violations

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Nov 2026

Historical Context: Statistical Intelligence in WWII

Objective: Infer the unknown total number of tanks produced, N, using a small sample of observed serial numbers.

1. Espionage Inflation

Conventional military intelligence relied on spy networks and captured prisoners, drastically overestimating production by **4.5–8.2**$\times$ (1,000–1,550 tanks/mo).

2. Order Statistics

Allied statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–50), mathematically estimating production at **169–327** tanks/mo.

3. Speer Ministry Ledgers

Post-war Albert Speer production records confirmed statistical estimates converged within **4%–10%** of actual production.

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	**169**	122	+38.5%
June 1941 (Panzer III/IV)	1,550	**244**	271	−**9.9**%
August 1942 (Panzer III/IV)	1,550	**327**	342	−**4.4**%

Core Inferential Objective: *Infer the unknown population ceiling N, moving beyond point estimates to complete uncertainty quantification.*

Mathematical Formulation: Data, Parameter, and Likelihood

Explicitly separating what is observed from the unknown parameter we seek to infer.

The Data-Generating Mechanism

Serials drawn by simple random sampling without replacement:

$$S_1, \ldots, S_k \overset{\text{SRSWOR}}{\sim} \{1, 2, \ldots, N\}$$

- **Observed Data:** $D = \{s_1, \ldots, s_k\}$ with sample size k.
- **Order Statistics:** $1 \leq s_{(1)} < s_{(2)} < \cdots < s_{(k)} = m$.
- **Sample Maximum:** $m = \max(s_1, \ldots, s_k)$.
- **Physical Boundary:** $N \geq m$ is an absolute physical constraint.

Inferential Target & Unordered Likelihood

- **Inferential Target:** The target is the unknown population ceiling $N \in \mathbb{N}_{\geq m}$, *not the individual serial numbers*.
- The serial numbers D are observed data; N is the unknown quantity we want to learn about.

$$P(D \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m)$$

- Likelihood is strictly zero for all $N < m$.
- Sample maximum m captures all sample dependence on N.

Key Insight: *The serial numbers are observed evidence; the population ceiling N is the unknown inferential target.*

Classical Solution: The UMVUE & Its Limitations

Frequentist point estimation achieves repeated-sampling unbiasedness but provides no posterior distribution.

The UMVUE Estimator

By the likelihood ratio criterion for minimal sufficiency (Bahadur, 1954), $M = \max_i S_i$ is complete and minimal sufficient.

$$\widehat{N}_{\mathrm{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1$$

$$\mathrm{Var}(\widehat{N}_{\mathrm{UMVUE}}) = \frac{(N-k)(N+1)}{k(k+2)}$$

- Unbiased across hypothetical repeated samples ($\mathbb{E}[\widehat{N}] = N$).
- Simple closed-form computation requiring only m and k.

Fundamental Limitations of Classical Approach

- **No Posterior Uncertainty Distribution:** The classical approach constructs a point estimator that is unbiased over repeated samples. It does not provide a posterior distribution describing uncertainty about N given this specific sample.
- **Boundary Violations:** Asymptotic Wald intervals ($\widehat{N} \pm 1.96 \cdot \mathrm{SE}$) extend below m, placing mass on physically impossible values ($N < m$).
- **Inflexible to Decision Costs:** Cannot adapt to asymmetric operational losses where underestimation carries disastrous strategic costs.

Transition to Bayes: *Point estimators provide only an average; Bayesian inference constructs the full posterior distribution $P(N \mid D)$.*

SECTION 01

Theoretical Properties & Bayesian Foundations

Inferring the posterior distribution, reference priors, loss geometry, and boundary admissibility

Bayesian Inference: What Are We Trying to Infer?

Establishing the inferential objective: moving from raw identifiers to an epistemic probability distribution.

1. What Do We Know?

- The observed sample of serial numbers $D = \{s_1, \ldots, s_k\}$.
- The sample maximum $m = \max(s_1, \ldots, s_k)$.
- Physical fact: the population must satisfy $N \geq m$.

2. What Is Unknown?

- The true population ceiling $N \in \{m, m+1, m+2, \ldots\}$.
- We do not know how far above m the true ceiling lies.

3. What Does Bayesian Inference Give Us?

A posterior distribution describing the probability of every possible population size, conditional on the observed data:

$$\boxed{P(N = n \mid D)}$$

4. What Do We Do With This Distribution?

- Quantify uncertainty via exact credible intervals.
- Derive optimal point estimates tailored to specific loss functions.
- Make principled decisions accounting for risk asymmetry.

Inferential Pipeline:

Observed Data $D = \{s_1, \ldots, s_k\}$ $\longrightarrow$ **Posterior Distribution** $P(N = n \mid D)$ $\longrightarrow$ Estimation & Decision Rules $\widehat{N}_L$

Core Message: *The posterior distribution itself is the primary inferential output; point estimates are derived afterwards.*

Ingredients of Bayesian Inference: Likelihood, Prior, Posterior

How observed evidence combines with prior assumptions to quantify uncertainty about N.

1. Likelihood: Data Evidence

What does the sample tell us about N?

$$P(D \mid N = n) = \frac{1}{\binom{n}{k}} \mathbb{I}(n \geq m)$$

- Measures the probability of observing D under each candidate n.
- Equals **zero for** $n < m$: automatically assigns zero posterior probability to physically impossible population sizes.

2. Prior: Initial Assumptions

What do we assume about N before observing data?

- **Flat Prior ($\pi \propto 1$):** Assumes equal prior weight across all candidate integers.
- **Reciprocal Prior ($\pi \propto 1/n$):** Discrete analog of scale invariance; assigns equal mass to multiplicative decades/octaves ($\sum_{A}^{cA} \frac{1}{n} \approx \log c$).

3. Posterior: Updated Belief

What do we believe about N after observing data?

$$P(N = n \mid D) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- Combines serial evidence with prior structure.
- Quantifies epistemic uncertainty across all physically admissible $n \geq m$.

Posterior Interpretation: *The posterior combines sample data with prior assumptions to assign exact probabilities to all candidate population sizes.*

Discrete Posteriors: Propriety & Moment Existence Thresholds

Different prior assumptions produce different posterior distributions, even when the observed data remain unchanged.

Exact Discrete Posterior PMFs ($n \geq m$)

- **Flat Prior ($\pi \propto 1$):**

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}$$

Normalizing constant telescopes exactly via negative hypergeometric summation.

- **Reciprocal Prior ($\pi \propto 1/n$):**

$$P(N = n \mid m, k) = \frac{1}{Z(m,k)} \frac{1}{n\binom{n}{k}}$$

$Z(m,k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}$ converges for all $k \geq 1$ (no simple closed form; evaluated numerically).

Moment Existence Thresholds & Pareto Limit

Prior Distribution	Proper Posterior	Finite Mean $\mathbb{E}[N]$	Finite Var $\mathrm{Var}(N)$
Flat Prior ($\pi \propto 1$)	$k \geq 2$	$k \geq 3$ ($\mathbb{E} = 2m - 2$)	$k \geq 4$
Reciprocal Prior ($\pi \propto 1/n$)	$k \geq 1$	$k \geq 2$ ($\mathbb{E} \approx 2m$)	$k \geq 3$

Continuous Pareto Limit ($N \gg k$):

$$p(N \mid m, k) \approx \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative CDF: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\widehat{N}_{\mathrm{med}} \approx m \cdot 2^{1/(k-1)}$.

Theoretical Takeaway: *Reciprocal prior guarantees posterior propriety for all $k \geq 1$, whereas flat prior requires $k \geq 2$ for propriety and $k \geq 3$ for finite mean.*

Estimating N from a Loss Function: Bayesian Decision Theory

A posterior distribution describes uncertainty, but a decision requires us to select a single estimate.

The Decision-Theoretic Framework

- The optimal estimate depends on the cost assigned to different estimation errors.
- Let $a \in \mathbb{N}_{\geq m}$ denote an estimation action.
- Loss function $L(N, a)$ quantifies the consequence of predicting a when true population is N.
- **Posterior Expected Loss (Risk):**

$$R(a) = \mathbb{E}[L(N, a) \mid D] = \sum_{n=m}^{\infty} L(n, a) P(N = n \mid D)$$

- **Optimal Bayes Estimator:**

$$\widehat{N}_L = \arg\min_a R(a)$$

Example: Squared-Error Loss (L_2)

When estimation errors are penalized quadratically:

$$L(N, a) = (N - a)^2$$

$$R(a) = \mathbb{E}[(N - a)^2 \mid D] = \mathrm{Var}(N \mid D) + (\mathbb{E}[N \mid D] - a)^2$$

Setting the derivative to zero yields:

$$\boxed{\widehat{N}_{L_2} = \mathbb{E}[N \mid D] \quad \text{(Posterior Mean)}}$$

- The posterior mean is not "the" Bayesian estimate; it is optimal specifically for squared-error loss.
- If errors carry asymmetric costs, a different estimator is required.

Core Rule: *Decisions follow inference: point estimators cannot be chosen without defining a loss function.*

Loss Functions & Bayesian Estimators: Matching Risk to Mission

Point estimators shift systematically with the decision-maker's loss geometry.

1. 0–1 Loss

$L(N, a) = \mathbb{I}(a \neq N)$

$\widehat{N} = \text{Mode} = m$

- Selects most probable single value.
- Severe negative bias (-16.7% at $k = 5$).

2. Absolute (L_1)

$L(N, a) = |N - a|$

$\widehat{N} = \text{Median}$

- Linear penalty on deviations.
- Robust against heavy Pareto tails (-0.9% bias).

3. Squared (L_2)

$L(N, a) = (N - a)^2$

$\widehat{N} = \mathbb{E}[N \mid D]$

- Quadratic penalty on large errors.
- Pulled rightward by skewed Pareto tail.

4. Check Loss L_q

$L_q = (N - a)(q - \mathbb{I}(N < a))$

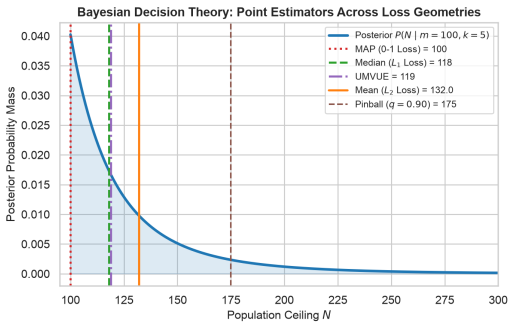
$\widehat{N} = F^{-1}(q \mid D)$

- $q = 0.5$: symmetric median.
- $q = 0.9$: undercounting penalized $9\times$ worse $\Longrightarrow$ defensive safety buffer.

Military Application: *In defense intelligence, underestimating enemy armor risks catastrophic defeat; asymmetric check loss ($q = 0.90$) provides a principled operational buffer.*

Numerical Illustration: Sensitivity of Estimates to Loss

Illustration using the flat-prior posterior, with sample maximum $m = 100$ and sample size $k = 5$.



Point Estimates Across Objectives ($m = 100, k = 5$)

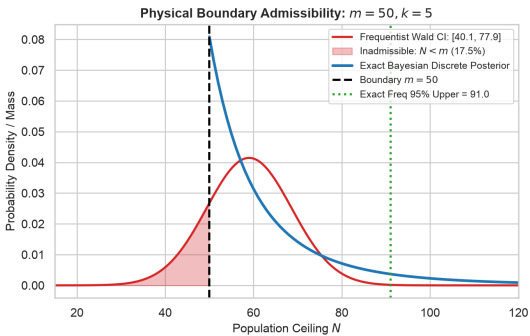
Objective / Loss	Estimator	Estimate	Strategic Meaning
0–1 Loss	MAP / Mode	**100**	Extreme lower bound; severe undercount
L_1 Absolute Loss	Posterior Median	**118**	Minimizes expected absolute error
L_2 Squared Loss	Posterior Mean	**132**	Quadratic balance; tail-inflated
Check Loss ($q = 0.90$)	90th Percentile	**175**	+75% defensive safety buffer

- Under the same observed data ($m = 100, k = 5$), optimal estimates span from 100 to 175.
- Posterior expected loss is minimized at a different point for each loss function.

Key Finding: *The same observed data can produce vastly different optimal estimates depending on how we value overestimation and underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian Intervals

The physical constraint $N \geq m$ must be respected by any valid uncertainty interval.



Frequentist Wald Boundary Collapse

Wald interval $\widehat{N} \pm 1.96 \cdot \mathrm{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$. Asymptotic Gaussianity fails completely near the support boundary.

Approximate Frequentist Pivotal CI

Large-N pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the physical boundary by construction.

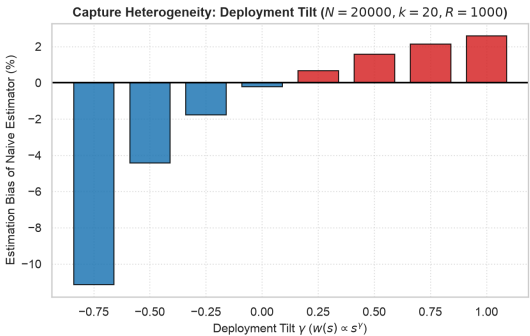
Bayesian Credible Coherence

Reciprocal-prior 95% credible interval strictly enforces physical admissibility ($P(N < m \mid D) = 0$) and approximately matches the pivotal bound $[50.0, 91.0]$.

Key Finding: *The reciprocal-prior posterior respects the physical lower bound and produces credible intervals that approximately agree with the large-N frequentist pivotal interval.*

Sensitivity to Non-Uniform Sampling: When the Maximum Misleads

Evaluating estimator robustness when capture probability varies with serial age: $w(s) \propto s^{\gamma}$.



Weighted Capture Mechanics

- $\gamma > 0$ (newer tanks prioritized): pushes $m \to N$, inducing $+1.6\%$ bias at $\gamma = +0.5$.
- $\gamma < 0$ (survivorship / depot attrition): induces -4.4% bias at $\gamma = -0.5$ and -11.1% bias at $\gamma = -0.75$.

Sufficiency Structure Breakdown

- **Known γ:** m remains sufficient for N via tilted factor $g_N(m; \gamma)$.
- **Unknown γ:** m alone ceases to be sufficient; joint minimal sufficient statistic is $(m, \sum_{i=1}^{k} \log s_i)$.

Methodological Warning: *Applying standard estimators when sampling is tilted introduces systematic bias; joint inference over (N, γ) is required.*

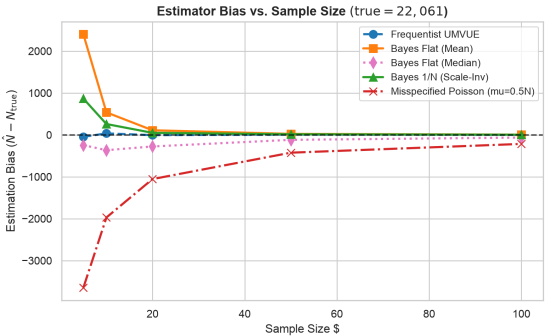
SECTION 02

Controlled Monte Carlo Simulation Study

Evaluating bias, RMSE contraction, interval calibration, and prior sensitivity across $R = 4,000$ trials ($N = 20,000$)

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$.



Frequentist UMVUE Unbiasedness

$\widehat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all k (-48.0 ± 54.6 at $k = 5$, $+5.2 \pm 6.0$ at $k = 50$). Unbiased over hypothetical repeated samples.

Bayes Reciprocal Prior

Posterior Mean has modest bias ($+767.0 \pm 56.8$ at $k = 5$), rapidly decaying to $+13.2$ at $k = 50$. Posterior Median exhibits negative shrinkage (-901.5 at $k = 5$).

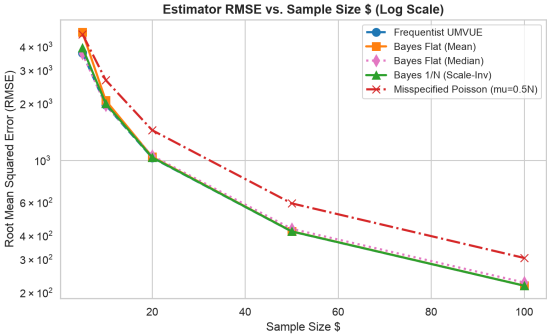
Flat Prior Tail Inflation

Skewed Pareto tail inflates flat posterior mean by $+2,168.6 \pm 60.7$ at $k = 5$.

Core Distinction: *Unbiasedness is a repeated-sampling frequentist property; Bayesian posterior summaries are chosen using the posterior distribution and a decision objective.*

Faster-Than-Standard-CLT Error Contraction: RMSE & Variance

Boundary order-statistic structure accelerates Root Mean Squared Error decay to $\mathcal{O}(k^{-1})$.



Theoretical Error Rate

Because N bounds the support, the UMVUE variance satisfies:

$$\mathrm{Var}(\widehat{N}) = \frac{(N-k)(N+1)}{k(k+2)} = \mathcal{O}(k^{-2})$$

giving accelerated rate RMSE $= \mathcal{O}(k^{-1})$, far outperforming standard CLT rate $\mathcal{O}(k^{-1/2})$.

Empirical Concordance

A $10\times$ increase in k ($5 \to 50$) yields a $\mathbf{9.0}\times$ error reduction ($3,455.2 \to 382.5$), closely matching theoretical $8.6\times$ factor and asymptotic $10\times$ scaling.

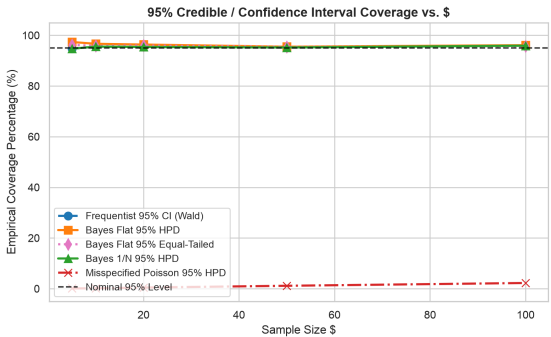
Estimator Convergence

For $k \geq 50$, all point estimators achieve identical precision (≈ 383).

Contraction Rule: *The order-statistic boundary structure drives an accelerated $\mathcal{O}(k^{-1})$ RMSE decay, reducing error by nearly an order of magnitude as sample size increases tenfold.*

Prior Sensitivity: Calibration and the Informative Prior Trap

Evaluating interval reliability and the catastrophic consequences of dogmatic prior misspecification.



Reciprocal Prior Calibration

Empirical coverage closely tracks nominal 95%: **94.4%** $\pm$ **0.36%** at $k = 5$ and **95.6%** $\pm$ **0.32%** at $k = 50$.

Informative Boundary Trapping

A dogmatically narrow misspecified Poisson($\mu = 10k$) prior traps posterior against the boundary $[m, m + 5]$, collapsing coverage to **0.3%**.

Hierarchical Hyperprior Resolution

Gamma hyperprior over Poisson rate ($\mathrm{CV} \geq 1$) provides adaptive dispersion, restoring coverage to **94.6%–95.0%** even under 50% prior mean error.

Calibration Finding: *The reciprocal prior produces empirical coverage close to 95% in our simulation. A misspecified Poisson prior causes severe collapse, while a hierarchical prior restores calibration.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Multi-dimensional evaluation across $R = 4,000$ replications on $N_{\text{true}} = 20,000$ (Seed 42).

Framework	Estimator	k	Mean Bias $\pm$ SE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Approximate Pivotal CI	Large-N Pivotal Bound	5	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 60.7$	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.1	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal $(1/n)$	Posterior Mean	5	$+767.0 \pm 56.8$	3,671.1 / 3,618.6	**94.4%** $\pm$ 0.36%	13,622
Bayes Reciprocal $(1/n)$	Posterior Median	5	-901.5 ± 52.3	**3,427.5** / —	**94.4%** $\pm$ 0.36%	13,622
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	50	$+5.2 \pm 6.0$	382.5 / 391.8	95.1% $\pm$ 0.34% (Wald)	1,536
Approximate Pivotal CI	Large-N Pivotal Bound	50	—	—	**95.6%** $\pm$ 0.32%	1,211
Bayes Reciprocal $(1/n)$	Posterior Mean	50	$+13.2 \pm 6.0$	382.8 / 392.0	**95.6%** $\pm$ 0.32%	1,209
Bayes Flat Prior	Posterior Mean	50	$+21.6 \pm 6.1$	383.4 / 392.4	**95.9%** $\pm$ 0.31%	1,237

1. Point Accuracy

UMVUE is exactly mean-unbiased; Bayes reciprocal median achieves lowest small-sample RMSE $(3,427.5)$.

2. Interval Reliability

Reciprocal credible interval achieves 94.4% coverage; Wald interval suffers from boundary distortion (90.0%).

3. Interval Precision

Reciprocal intervals are substantially sharper $(13,622)$ than flat prior intervals $(18,535)$.

Synthesis: *The reciprocal prior provides a useful balance between estimation accuracy, empirical calibration, and physical admissibility in the simulated setting.*

SECTION 03

Real-World Empirical Stress-Testing

Evaluating textbook assumptions against WWII archives and 541,909 commercial transactions

Empirical Stress-Testing: Two Real-World Datasets

Examining what happens when textbook assumptions meet physical operational systems.

1. Archival WWII German Tank Ledgers

- **Data Source:** Waffenamt ordnance ledgers and Speer Ministry production books.
- **Identifiers Analyzed:** Chassis serial blocks and sequential ZF gearbox stampings.
- **Operational Reality:** Pre-allocated factory blocks; chassis diverted to assault guns (StuG III/IV).

2. Commercial Invoice Ledger (Online Retail)

- **Data Source:** 541,909 transaction line items across 4,372 customer accounts.
- **Serials Analyzed: 22,061 distinct sequential sales invoices**.
- **Sequence Address Span:** $[536365, 581587] \implies \mathbf{45,223}$ slots.
- **Integer Voids:** 23,162 unrecorded slots (**51.2% void rate**).

Crucial Methodological Clarification: These datasets are used to examine the **limitations of sequential-number inference in operational settings**, NOT to claim that both datasets satisfy the German Tank sampling model.

Distinction: *541,909 represents raw transaction line items; 22,061 represents distinct sales invoice serial numbers.*

Finding 1: Chronological Generation vs. Uniform Sampling

Real-world serials are generated dynamically over time, not drawn as static random samples.

What Was Measured & Calculated

- **Measured:** Arrival timestamps and sequential numeric identifiers across all recorded transactions.
- **Calculated:** Weekly arrival volume $\Delta s / \Delta t$ and cumulative serial growth trajectories.
- **Observed Stream Dynamics:** Weekly volume accelerates from $\approx$ **370** invoices/week in Q1 to $>$ **790** invoices/week in Q4, peaking at **894** invoices/week in mid-November.

Research Question Impact & Statistical Conclusions

- **Textbook Assumption:** Assumes a *static population* from which observations are sampled randomly.
- **Operational Reality:** Invoices form an open, chronological transaction stream with non-stationary volume.
- **Exchangeability Violation:** Chronological ordering induces strong temporal autocorrelation. Successive observations are not exchangeable in the way the textbook model assumes.

Statistical Conclusion: *A static population-ceiling model cannot automatically be applied to a dynamic transaction stream.*

Finding 2: Integer Voids & System Architecture

Operational workflows generate massive integer voids (51.2% unrecorded rate).

Measured Sequence Breakdown

- **Total Integer Address Slots: 45, 223** ($s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1$).
- **Recorded Sales Invoices: 22, 061** (48.78% recorded).
- **Unrecorded Slots (Voids): 23, 162** (**51.22**% void rate).
- **Gap Distribution:** Median gap $\Delta s = 1.0$, mean 2.05, max 76.

Verified Facts vs. Operational Hypotheses

- **Verified Cancellations (3,836):** Exactly 3,836 C-prefixed invoice numbers exist in the raw dataset, directly explaining **16.56% of voids**.
- **Operational Hypotheses (19,326):** Without internal server logs, the remaining 83.44% voids plausibly reflect uncommitted checkout sessions, gateway dropouts, or distributed sequence caching.
- **Wartime Parallel:** Waffenamt pre-allocated chassis blocks; assembly diversions created sequence gaps.

Crucial Distinction: *The 3,836 cancellations are verified in data; the remaining 19,326 voids are plausible operational hypotheses, not verified findings.*

Finding 3: Goodness-of-Fit & The Uniformity Fallacy

Uniform dispersion across an address span is not exchangeability.

What Was Measured & Calculated

- Normalized span positions $u_i = (s_i - s_{\min})/(s_{\max} - s_{\min})$ evaluated against uniform CDF $F(u) = u$.
- **Maximum Absolute Discrepancy:** $D = \sup_u |F_n(u) - u| = 0.012$ $(n = 22,061)$.
- **Invalidity of Standard KS Table:** Estimating endpoints from data pins $u_{(1)} = 0$ and $u_{(n)} = 1$. Textbook KS distributions do not apply; formal testing requires parametric bootstrap.

What the ECDF Tells Us vs. What It Does Not

What the ECDF Tells Us	What It Does NOT Tell Us
Even distribution across span	Whether observations are exchangeable
Absence of massive block gaps	Whether sampling mechanism is uniform
Descriptive numerical regularity	Validity of the German Tank model

- A linear ECDF merely shows even allocation across the year.
- Successive observations are strongly autocorrelated in time.

Methodological Rule: *Never interpret a non-significant goodness-of-fit test as proof of exchangeable uniform sampling.*

Finding 4: Estimand Disambiguation & Thinning Models

What exactly are we trying to count? Sequence capacity vs. realized transaction volume.

Estimand Disambiguation

- **Estimand A ($N_{\text{span}} = 45,223$):** Total allocated address capacity ($s_{\max} - s_{\min} + 1$). Raw estimator without base offset yields $\approx 593k$.
- **Estimand B ($N_{\text{count}} = 22,061$):** Net volume of realized, active completed items. Shifting by $s_{\min}$ gives $+105\%$ error over active count.
- **Oracle Trap:** Rank re-indexing requires knowing all valid keys in advance (un-deployable in live audits).

The Bernoulli Thinning Model & Caveats

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho)$$

- **Retrospective Calculation:** Using the complete dataset ($k = 22,061$), $\widehat{\rho}_{\text{retrospective}} = \frac{22,060}{45,222} \approx 0.4878$.
- **Operational Identifiability Caveat:** This calculation uses the complete observed invoice set. In an incomplete field sample ($k \ll N$), the sample spacing ratio estimates sample density, not ρ.
- Estimating both N_{span} and ρ from an incomplete sample requires auxiliary data.

Conceptual Takeaway: *Bernoulli thinning clarifies the estimand distinction conceptually; estimating both capacity and void rate requires auxiliary identifying structure.*

Connections to Other Areas of Statistical Inference

How discrete order-statistic inference connects across domains, and where its boundaries lie.

Conceptual Cross-Domain Connections

- **Dynamic Production:** Monotonic state-space models ($N_t \geq N_{t-1}$) enforce output non-negativity across periods.
- **Software Reliability:** Jelinski–Moranda defect exhaustion models mirror serial discovery for software release readiness.
- **Aviation Safety:** Beta–Bernoulli mixtures detect cloned counterfeit serial numbers in critical parts supply chains.
- **Wildlife Abundance:** Two-stage cascades infer tag loss before estimating population size via mark-recapture.

Methodological Boundaries

- **Non-Sequential Keys:** Randomized UUIDv4 or cryptographic hashes completely destroy order-statistic sufficiency.
- **Non-Random Gaps:** Classified blocks or unrecorded overseas entities require informative gap priors.
- **Dynamic Streams:** Fast-changing arrival rates require time-varying Poisson arrival processes.
- **Identifiability Limits:** Incomplete samples cannot simultaneously identify capacity and void rate without auxiliary data.

Synthesis: *Order-statistic inference generalizes broadly across domains, provided analysts explicitly model data-generating mechanisms and respect identifiability boundaries.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

The target is the discrete ceiling N, with evidence captured by sample maximum m. Scale-invariant reciprocal prior $\pi(N) \propto 1/n$ yields proper posteriors ($k \geq 1$), respects physical boundaries ($N \geq m$), and matches the large-N pivotal bound.

2. Decision Theory & Rates

Point estimation follows loss geometry: median minimizes absolute error; check loss ($q = 0.9$) builds defensive buffers. Extreme-value order statistics produce accelerated $\mathcal{O}(k^{-1})$ RMSE contraction ($9\times$ reduction for $10\times$ sample).

3. Real-World Reality

Operational data violate textbook assumptions via chronological streaming and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction volume.

Central Research Answer: Bayesian inference provides a principled framework for quantifying uncertainty about an unknown discrete population ceiling, selecting estimates under mission-specific decision objectives, and evaluating inferential reliability. Its practical validity in the physical world, however, depends entirely on whether operational data-generation mechanisms satisfy model assumptions.

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